KATERINA TANG

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Education

Cornell University, Ithaca, NY
Ph.D. in Applied Mathematics *Expected Summer 2027*

- Committee: C. Lindsay Anderson, Vivek Srikrishnan, and David Bindel

Kenyon College, Gambier, OH
B.A. in Mathematics, minor in German *May 2021*

- Valedictorian, *summa cum laude*

Publications

** denotes equal author contribution*

- **K. Tang***, D. Kaiser*, W. Thompson*, J. G. Young, L. Hébert-Dufresne, N. W. Landry. "Emergent contagion complexity: Disentangling mechanistic complexity from correlated heterogeneity," *under review*. arXiv:2607.25139.
- **K. Tang**, V. Srikrishnan, and J. Kulik. "Bayesian inference of hypergraph structure from scarce and noisy dynamical observations," *Journal of Physics: Complexity*, 2026. DOI: 10.1088/2632-072X/aeae02.
- **K. Tang**, M. V. Liu, C. L. Anderson, and V. Srikrishnan. "Identification of pressure points in modern power systems using transfer entropy," *Cell Reports Sustainability*, 2026. DOI: 10.1016/j.crsus.2026.100660.

Teaching

Cornell University

- **Graduate Teaching Assistant**, Math 1106 Modeling with Calculus for the Life Sciences
  Fall 2026
- **Graduate Teaching Assistant**, Math 1900 Introduction to Calculus for Engineers
  Summer 2026
- **Head Teaching Assistant**, Math 1920 Multivariable Calculus for Engineers
  Fall 2025, Spring 2026
- **Graduate Teaching Assistant**, Math 1920 Multivariable Calculus for Engineers
  Fall 2021, Spring 2022, Fall 2022, Fall 2024, Spring 2025, Fall 2025, Spring 2026

Kenyon College

- **Grader**, Math 224 Linear Algebra
  Fall 2020
- **Grader**, Math 213 Calculus III
  Fall 2018, Spring 2019, Fall 2019
- **Peer tutor**, Kenyon Student Accessibility and Support Services
  Spring 2018

Honors and Awards

Graduate Student Teaching Award, Cornell Math Department — Fall 2025

Reginald B. Allen Prize, Kenyon Math Department — Spring 2021
Awarded to a student who has shown unusual promise in mathematics

Phi Beta Kappa, member — Fall 2020

Barry M. Goldwater Scholarship — Spring 2020

Clare Boothe Luce Undergraduate Research Scholarship — 2019
Competitive undergraduate research scholarship for women in STEM

Pi Mu Epsilon, member — Fall 2018

Mentorship and Service

Cornell University

- **Co-director & mentor**, Center for Applied Mathematics First-Year Mentoring Program
  Aug 2024 – present
- **Graduate student mentor**, Cornell Chapter of the Association for Women in Mathematics
  Sep – Dec 2021

Kenyon College

- **Mentor**, KSTEM Mentoring Program
  Aug 2020 – May 2021
- **Founding member and mentor**, Kenyon College Chapter of the Association for Women in Mathematics
  Aug 2018 – May 2021
- **Volunteer & staff**, ATHENAS Saturday Science Program
  Jan – Dec 2019

Selected Presentations

** denotes poster presentation*

- "The emergent complexity of contagion," *Student Colloquium in Applied Mathematics*, Ithaca, NY, 2026.
- "Emergent complexity of contagion: Correlated heterogeneity in simple contagions mimics complex contagions," *NetSci*, Boston, MA (2026).
  *Received **Best Student Talk** award.*
- "Bayesian hypergraph inference from dynamics," *Student Colloquium in Applied Mathematics*, Ithaca, NY, 2025.
- "Identification of infrastructure pressure points in a decarbonized grid across climatic-technological scenarios," *Annual Conference of the Society for Decision Making Under Deep Uncertainty*, Denver, CO (2024).
- "Identification of infrastructure pressure points in a decarbonized grid across climatic-technological scenarios," *AGU Annual Meeting*, Washington, DC (2024).*

- "A clustering-based method for interpretable multi-class scenario discovery," *MultiSector Dynamics Workshop*, Davis, CA (2023).*
- "Existence and Character of Solutions to the Erdős-Straus Equation," *Joint Mathematics Meeting*, Denver, CO (2020).*
  *Received **MAA Outstanding Poster** award.*